

BUSINESS ACCOUNT SYSTEM

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Revision History

Revision	Date	Description	Modified by
1	2018/05/24	Initial version	Jimmy
1.1	2018/05/29	Modified API format : 1. Modified API names. 2. Modified API headers. API Add get_profile	Nathan Jimmy

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1. Introduction

This document covered APIs, webhook events and supported message types of Letstalk Business Accounts. There are two security levels, one can choose to integrate either ways to meet security requirements.

Level 1: Letstalk service will do the RSA for end-to-end encryption, the AES key will be stored in the Letstalk service, provides easy integration with basic security.

Level 2: Customers' bots take the responsibility to do end-to-end encryption by RSA, the Letstalk service will not be aware of any AES keys, this level provides advanced security.

Customers' bots should use AES to encrypt all the message content in both levels.

We will provide sandbox environment for integration test.

The necessary parameters for applying for an account include **Account Name**, **Password, Display Name**, **Security level**. We will give you **api_key**, **hash_key** and **hash_id**.

2. API

1.2. change_display_name

- To change the display name of business accounts.
- Name: **bot/change_display_name/{{user_name}}**
- Method: PUT
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
  “display_name”: String
}
```

- Output: JSON object

```
{
   "status": int
   "message": String
}"
```

- Status:

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Code	Description
0	failure
1	success

1.3. upload_avatar_image

- To upload avatar/portrait images in 640x640.
- Name: **bot/upload_avatar_image/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
  “image”: String (base64 string)
}
```

- Output: JSON object

```
{
   "status": int,
   "message": String
}
```

- Status:

Code	Description
0	failure
1	success

1.4. set_welcome_msg

- To setup welcome messages for new contact.
- Name: **bot/set_welcome_msg/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)

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}

- Input: JSON object

```
{
  “message”:String
}
```

- Output: JSON object

```
{
   "status": int,
   "message": String
}"
```

- Status:

Code	Description
0	failure
1	success

1.5. set_auto_reply

- To set a message auto replied to contact’s messages.
- Name: **bot/set_auto_reply/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
 “message”:String,
 “enable”: int (0:off 1:on)
}
```

- Output: JSON object

```
{
   "status": int,
   "message": String
}"
```

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- Status:

Code	Description
0	failure
1	success

1.6. push

- To send a message to contact.
- Name: **bot/push/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
  “receiver”: Array [receiver A, receiver B…] MAX:50
  “payload”:String
}
```

- Output: JSON object

```
{
  "status": int
  "message": String
}"
```

- Status:

Code	Description
0	failure
1	success

1.7. push_to_tag

- To send message to tagged contacts.
- Name: **bot/push_to_tag/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json

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Authorization : Bearer base64(api key)

}

- Input: JSON object

```
{
  “tag”:String
  “payload”:String
}
```

- Output: JSON object

```
{
   "status": int
   "message": String
}"
```

- Status:

Code	Description
0	failure
1	success

1.8. modify_tag

- To add or delete tags, number of tags are limited to 20.
- Name: **bot/modify_tag/{{user_name}}**
- Method: POST / DELETE
- Header: {

Content-Type : application/json

Authorization : Bearer base64(api key)

}

- Input: JSON object

```
{
  “tag”:String
 ~~“action”:int (1:add, 2:delete)~~
}
```

- Output: JSON object

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```
{
  "status": int
  "message": String
}
```

- Status:

Code	Description
0	failure
1	success

1.9. list_tag

- To list all tags
- Name: **bot/list_tag/{{user_name}}**
- Method: GET
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input:None
- Output: JSON object

```
{
  "status": int
  "message": String
  "tags": [String, ...]
}
```

- Status:

Code	Description
0	failure
1	success

1.10. get_contacts

- To get contacts by tags, list all contacts when tag is not provided.
- Name: **bot/get_contacts/{{user_name}}?tag=xxx**
- Method: GET
- Header: {

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Content-Type : application/json

Authorization : Bearer base64(api key)

}

- Input: none
- Output: JSON object

```
{
  "status": int,
  "message": String,
  “tag”:(String),
  "contacts":[
      {"user_id":String (Letstalk id),”avatar_url”:String, “display_name”:String},
      {"user_id":String (Letstalk id),”avatar_url”:String, “display_name”:String},
      ...
  ]
}
```

- Status:

Code	**Description**
0	failure
1	success

1.11. modify_contact_tag

- To tag or to remove the tag of some contact.
- Name: **bot/modify_contact_tag/{{user_name}}**
- Method: POST/DELETE
- Header: {

Content-Type : application/json

Authorization : Bearer base64(api key)

}

- Input: JSON object

```
{
  “contact”:String,
  “action”:int (1:Add 2:Delete)
  “tag”:String
}
```

- Output: JSON object

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```
{
  "status": int
  "message": String
}
```

- Status:

Code	Description
0	failure
1	success

1.12. set_hash_key

- Upload new AES key, if encryption level 2 does not allow uploaded.
- Name: **bot/set_hash_key/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
 “hash_key”: String
 “hash_id”:longlong
}
```

- Output: JSON object

```
{
  "status": int
  "message": String
}
```

- Status:

Code	Description
0	failure
1	success

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1.13. get_contact_public_key

- Get contact RSA public key. (Level 2 only)
- Name: **bot/get_contact_public_key/{{user_name}}?contact=xxxx**
- Method: GET
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: none
- Output: JSON object

```
{
  "status": int
  "message": String
  “public_key”:String
}
```

- Status:

Code	Description
0	failure
1	success

1.14. set_key_to_contact

- Set encryption key to contact. (Level 2 only)
- Name: **bot/set_key_to_contact/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
 “contact”: String,
 “encryption_key” : String
 “hash_id”:long
}
```

- Output: JSON object

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```
{
  "status": int
  "message": String
}
```

- Status:

Code	Description
0	failure
1	success

1.15. set_webhook

- Setup you webhook URL.
- Name: **bot/set_webhook/{{user_name}}**
- Method: POST
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
  “url”:String
}
```

- Output: JSON object

```
{
  "status": int
  "message": String
}
```

- Status:

Code	Description
0	failure
1	success

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1.16. get_profile

- Get contact profile information
- Name: **bot/get_profile/{{user_name}}**
- Method: GET
- Header: {
  Content-Type : application/json
  Authorization : Bearer base64(api key)
  }
- Input: JSON object

```
{
  “contact”: String
}
```

- Output: JSON object

```
{
  "status": int,
  "message": String,
  ”avatar_url”:String,
  “display_name”:String
}
```

- Status:

Code	Description
0	failure
1	success

3. Webhook Event

3.1. Common properties

- Description

Property	Type	Description
type	String	Identifier for the type of event.
timestamp	Number	Time of the event in milliseconds.

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source	Object	Source user object with information about the source of the event.

- Source User

Property	Type	Description
type	String	user
user_id	String	Letstalk ID of the source user.

3.2. Message Event

- Description

Property	Type	Description
type	String	message
payload	String	message payload. This will be an encrypted JSON String.

- Example

```
{
 “type”: “message”,
 “timestamp”: 1527151205790,
 “source”:{
       “type”:”user”,
       “user_id”:”jimmy.chen”
  },
  “payload”:String
}
```

3.3. Add contact Event

- Description

Property	Type	Description
type	String	add_contact

- Example

```
{
 “type”: “add_contact”,
 “timestamp”: 1527151205790,
 “source”:{
```

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```
        “type”:”user”,
        “user_id”:”jimmy.chen”
  }
}
```

3.4. Delete contact Event

- Description

Property	Type	Description
type	String	delete_contact

- Example

```
{
 “type”: “unsubscribe”,
 “timestamp”: 1527151205790,
 “source”:{
        “type”:”user”,
        “user_id”:”jimmy.chen”
  }
}
```

3.5. Ask Key Event

- Description

Property	Type	Description
type	String	ask_key

- Example

```
{
 “type”: “ask_key”,
 “timestamp”: 1527151205790,
 “source”:{
        “type”:”user”,
        “user_id”:”jimmy.chen”
  }
}
```

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4. Incoming Payload

4.1. Common properties

- Description

Property	Type	Description
type	String	Identifier for the type of msg. 0: Text, 2: Image, 3: Voice, 6: Video, 7: GIF, 8: File
user_id	String	Sender ID
receiver	String	Receiver ID
id	Number	Message ID
time	Number	Time of the Message in milliseconds.
send_time	Number	Time of the Message send in milliseconds.
due_date	Number	Time of the Message due date in milliseconds.
from	Number	From Device 0: Mobile, 1:Desktop
hash_content	String	Encrypted text.
hash_contetn_key	String	The plain text to SHA1
hash_id	Number	AES key id.
hash_type	Number	Encryption mode 4: AES

4.2. hash_content properties

- Text Type

Property	Type	Description
msg	String	Text content
quote	Object	Quote info

Quote Object

Property	Type	Description
display_name	String	User display name. Found in the quote object.

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annount_name	String	User Letstalk ID. Found in the quote object.
content	String	Text content. Found in the quote object.

- Image Type

Property	Type	Description
preview_image_url	String	Preview image URL
original_image_url	String	Image URL
zip_password	String	Zip file password

- Voice Type

Property	Type	Description
voice_url	String	URL of voice file
voice_duration	Number	Length of voice file.(milliseconds)
zip_password	String	Zip file password

- Video Type

Property	Type	Description
preview_image_url	String	URL of preview image
video_url	String	URL of video file
video_duration	Number	Length of video file.(milliseconds)
video_size	Number	Size of video file. (bytes)
zip_password	String	Zip file password

- GIF Type

Property	Type	Description
gif_key	String	Giphy key. (https://giphy.com/)

- File Type

Property	Type	Description
file_name	String	File name
file_url	String	URL of file
file_size	Number	Size of file. (bytes)

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zip_password	String	Zip file password

- Example:


```
{
  "id":350,
  "hash_content_key":"f922b4b70151303c1f85bd4e7712c0a033fa8ba7",
  "time":1527214433078,
  "type":0,
  "hash_type":2,
  "user_id":"jimmy100.chen",
  "hash_content":"x9CK80e2D2ulQX6+48AmzA==",
  "from":0,
  "send_time":1527214433177,
  "due_date":2788654433177,
  "receiver":"jimmy.chen",
  "hash_id":1526294541848
}
```


5. Outgoing Payload

5.1. Common properties

Property	**Type**	**Description**
type	String	Identifier for the type of msg. 60: Text, 61: Image, 62 : Image map
user_id	String	Sender ID
receiver	String	Receiver ID
id	Number	Message ID
time	Number	Time of the Message in milliseconds.
send_time	Number	Time of the Message send in milliseconds.
due_date	Number	Time of the Message due date in milliseconds.
from	Number	From Device 2: Business Account
hash_content	String	Encrypted text.

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hash_contetn_key	String	The plain text to SHA1
hash_id	Number	AES key id.
hash_type	Number	Encryption mode 4: AES

5.2. hash_content properties

- Text Type

Property	**Type**	**Description**
msg	String	Text content

- Image Type

Property	**Type**	**Description**
preview_image_url	String	Image URL **HTTPS** (Required) JPEG MAX: 1024 x 1024 MAX: 1MB
original_image_url	String	Preview image URL **HTTPS** (Required) JPEG MAX: 640 x 640 MAX: 1MB

- Image Map Type

Property	**Type**	**Description**
base_url	String	Image URL **HTTPS** (Required) JPEG or PNG Image width: 240px, 300px, 460px, 700px, 1040px MAX: 1MB
alt_text	String	Alternative text
base_size_width	Number	Width of base image (set to 1040px)
base_size_height	Number	Height of base image (set to the height that corresponds to a width of 1040px)

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actions	Array of Action object	Action when tapped, Max:5

- Action object

URI Action type

Property	Type	Description
type	String	uri
link_uri	String	Web page URL
area	Area object	Defined tappable area.

Message Action type

Property	Type	Description
type	String	message
text	String	Message to send MAX:400 characters
area	Area object	Defined tappable area.

- Area object

Property	Type	Description
x	Number	Horizontal position relative to the top-left corner of the area
y	Number	Vertical position relative to the top-left corner of the area
width	Number	Width of the tappable area
height	Number	Height of the tappable area

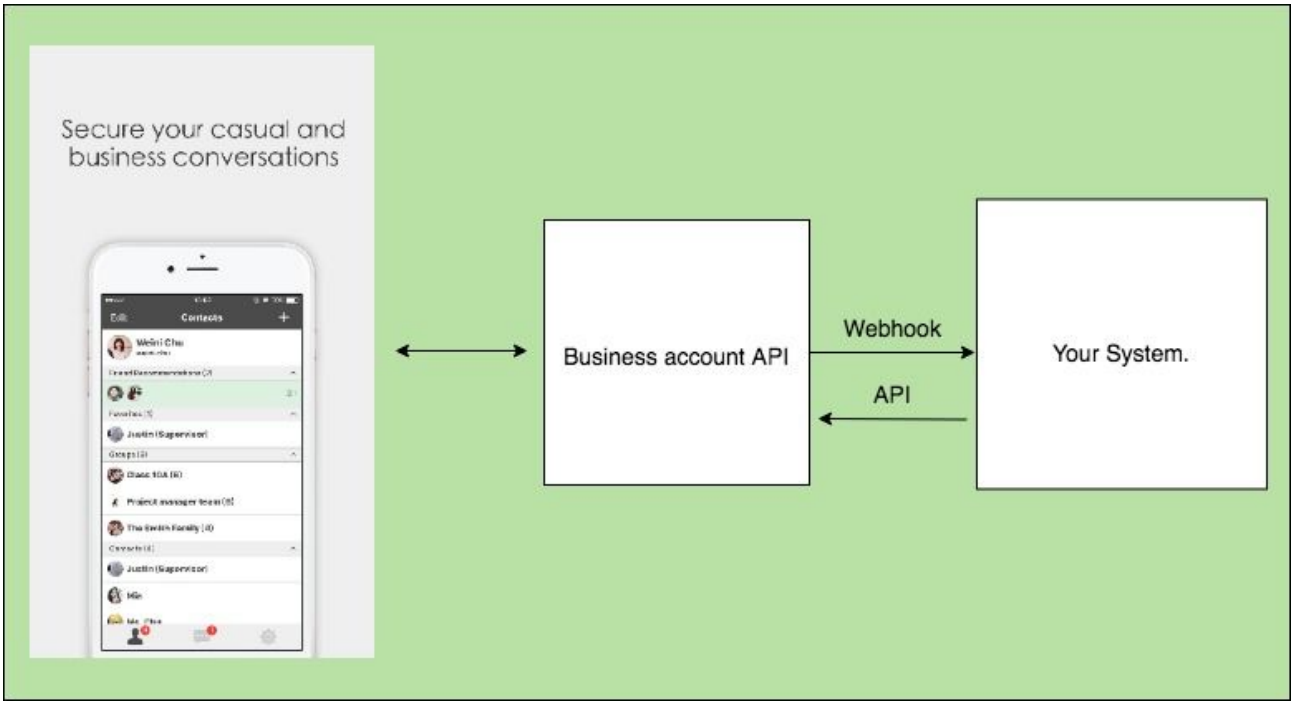
6. Guides

6.1. How it works

The Business Account API allows for data to be passed between the server of your application and the Lestalk Platform. When a user sends your business account a message, a webhook is triggered and the Letstalk Platform sends a request to your webhook URL. Your server then sends a request to the Lestalk Platform to respond to the user. Requests are sent

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over HTTPS in JSON format.



6.2. What you can do with Business Account API

1. Set up a welcome message when user add account to contact.
2. Set up an auto reply message. Can be used when your service is maintained.
3. Set up an avatar image.
4. Push message to contacts.
5. Push various message types. (ex: text, image, imagemaps)

6.3. How to send key to contact. When webhook receive Ask Key event.

1. Receive Ask Key event, you can get **user_id** from source object.

```
{
 “type”: “ask_key”,
 “timestamp”: 1527151205790,
 “source”:{
       “type”:”user”,
       “user_id”:”jimmy.chen”
  }
}
```

2. Use **get_contact_public_key** API, Get contact RSA public key.
3. Use contact RSA public key to encrypt business account hash_key with RSA 2048-bit encryption.
4. Use **set_key_to_contact** API, Set encryption_key to contact.

6.4. How to decrypt **hash_content**. From payload JSON String.

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1. Check **hash_type** is equal **4**.
2. Check **hash_id** is the same as your own.
3. Use your own **hash_key** to decrypt **hash_content** with AES 256-bit encryption.

6.5. How to encrypt **hash_content (Plain Text)**

1. Use your own **hash_key** to encrypt **hash_content(Plain Text)** with AES 256-bit encryption.

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